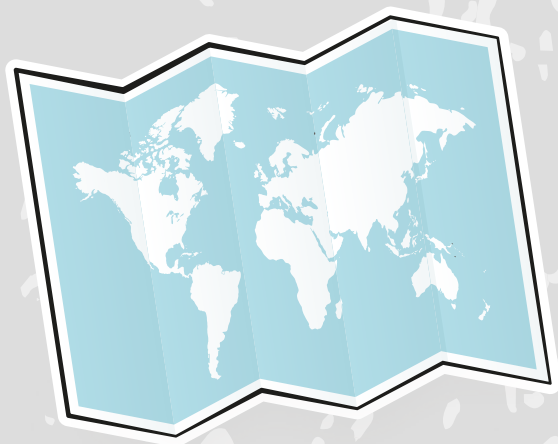
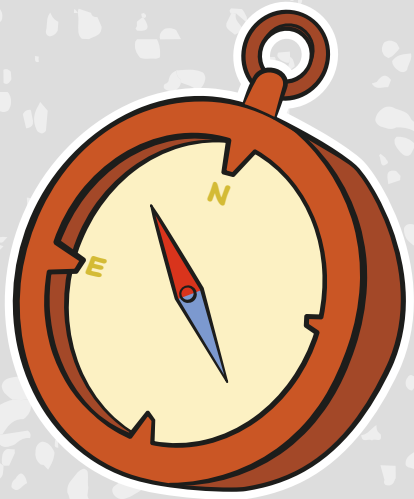
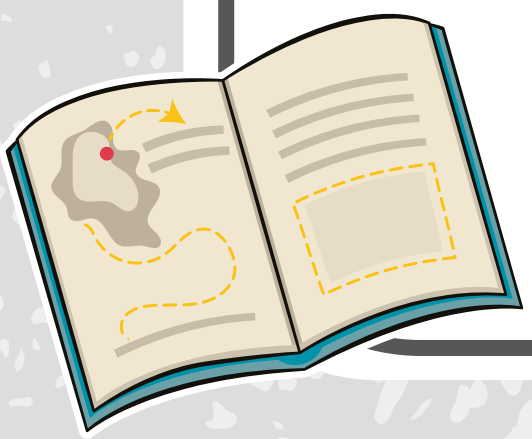


PROFESSOR
FERRETTO

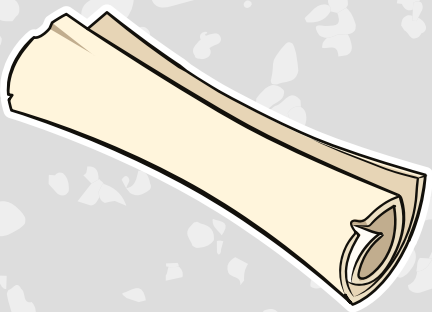


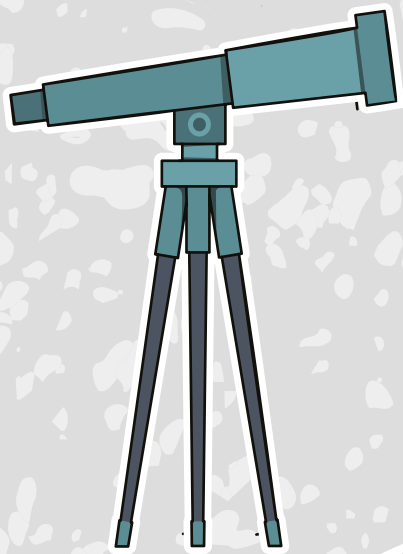
Geografia

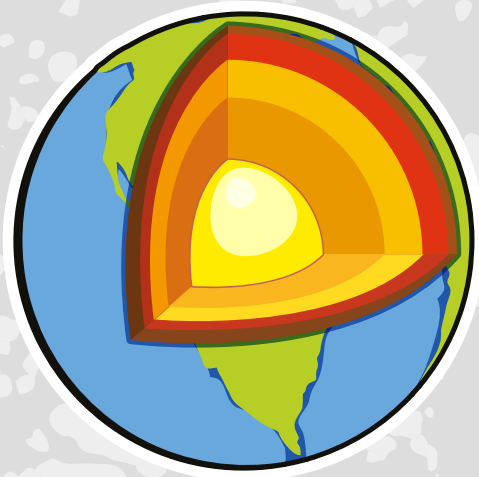
GEOLOGIA

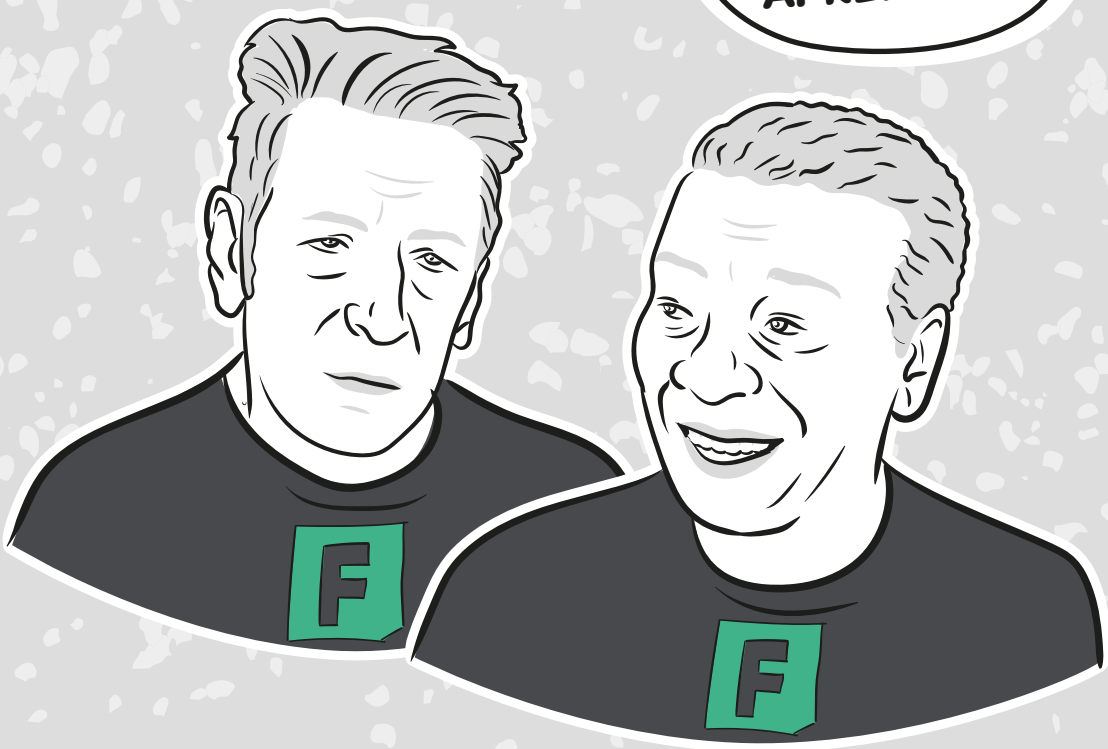


HOJE É UM ÓTIMO DIA PRA VOCÊ APRENDER!



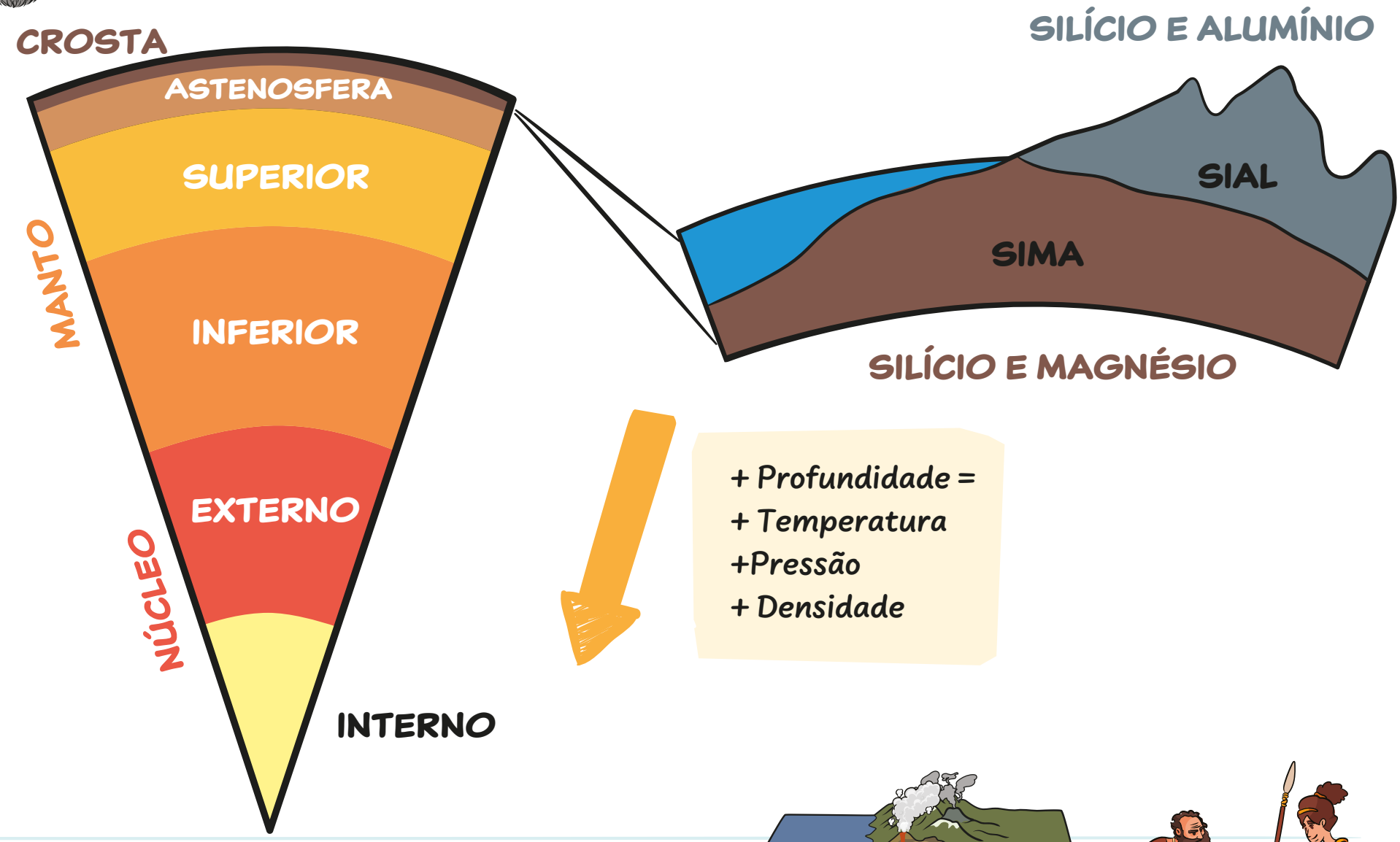






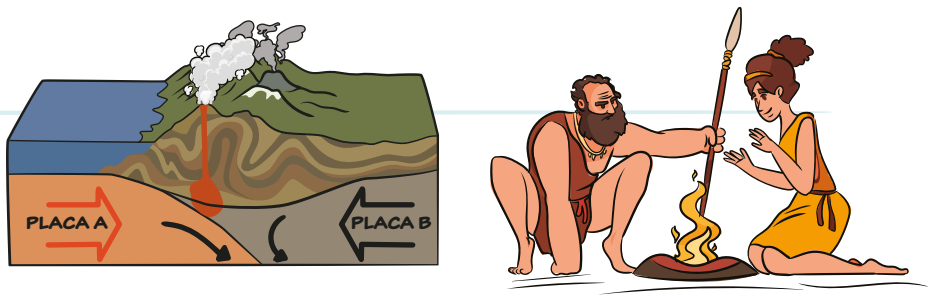
GEOLOGIA

ESTRUTURA INTERNA DA TERRA



CENOZOICA
(65 milhões de anos)

- QUATERNÁRIO (1,8)
- TERCIÁRIO (65)

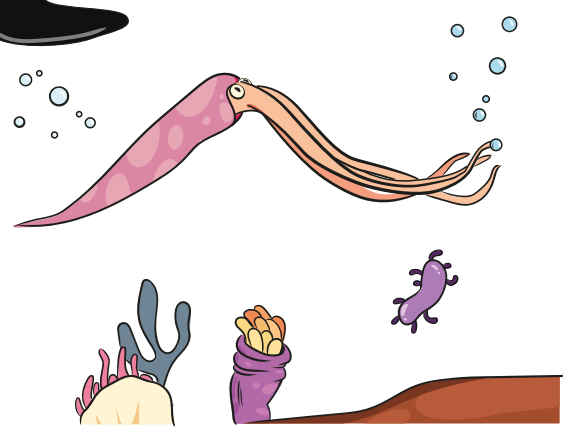


MESOZOICA
(251 milhões de anos)

CENOZOICA
MESOZOICA
PALEOZOICA

PALEOZOICA
(542 milhões de anos)

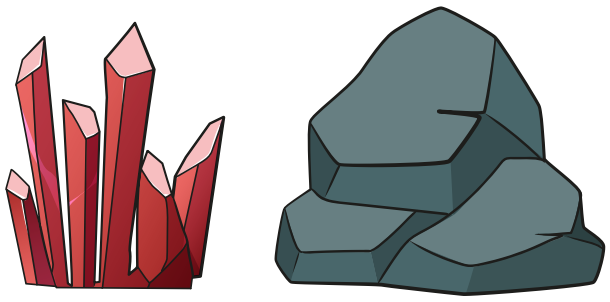
Reservas de carvão mineral



PRÉ-CAMBRIANA

PROTEROZOICO
(2,5 bilhões de anos)

Rochas e minerais metálicos



ARQUEOZOICO
(3,58 bilhões de anos)

Primeiras formas de vida

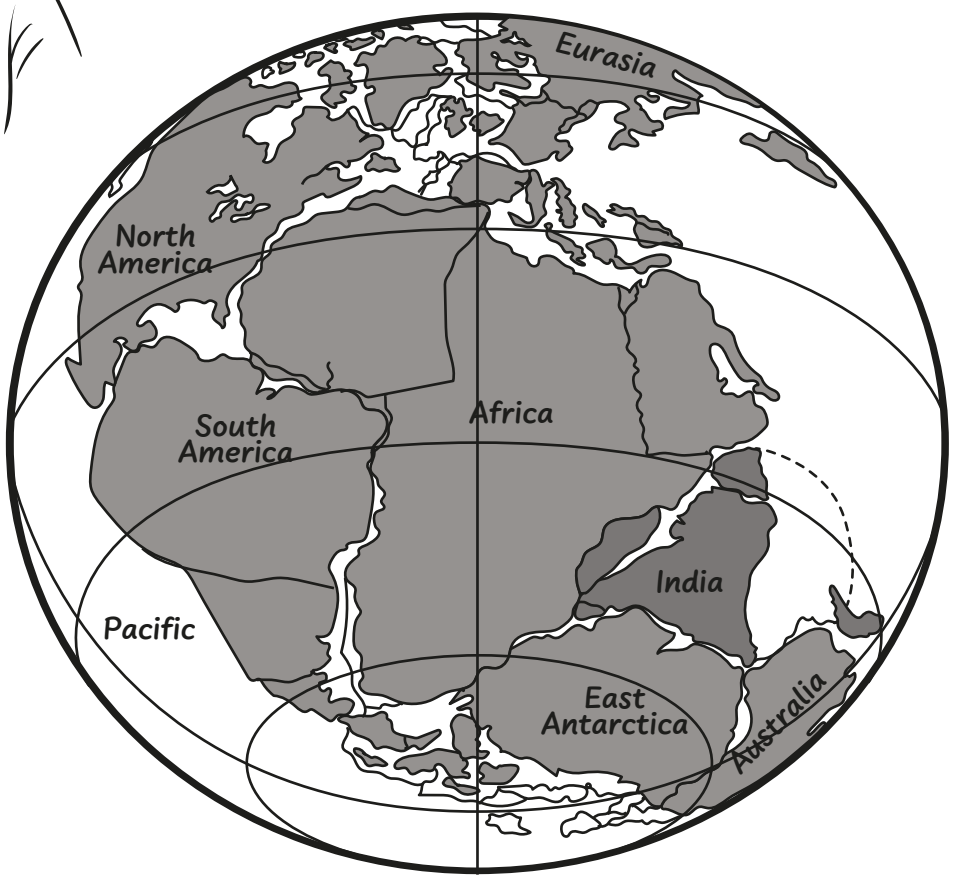
HADEANO
(4,56 bilhões de anos)

DERIVA CONTINENTAL

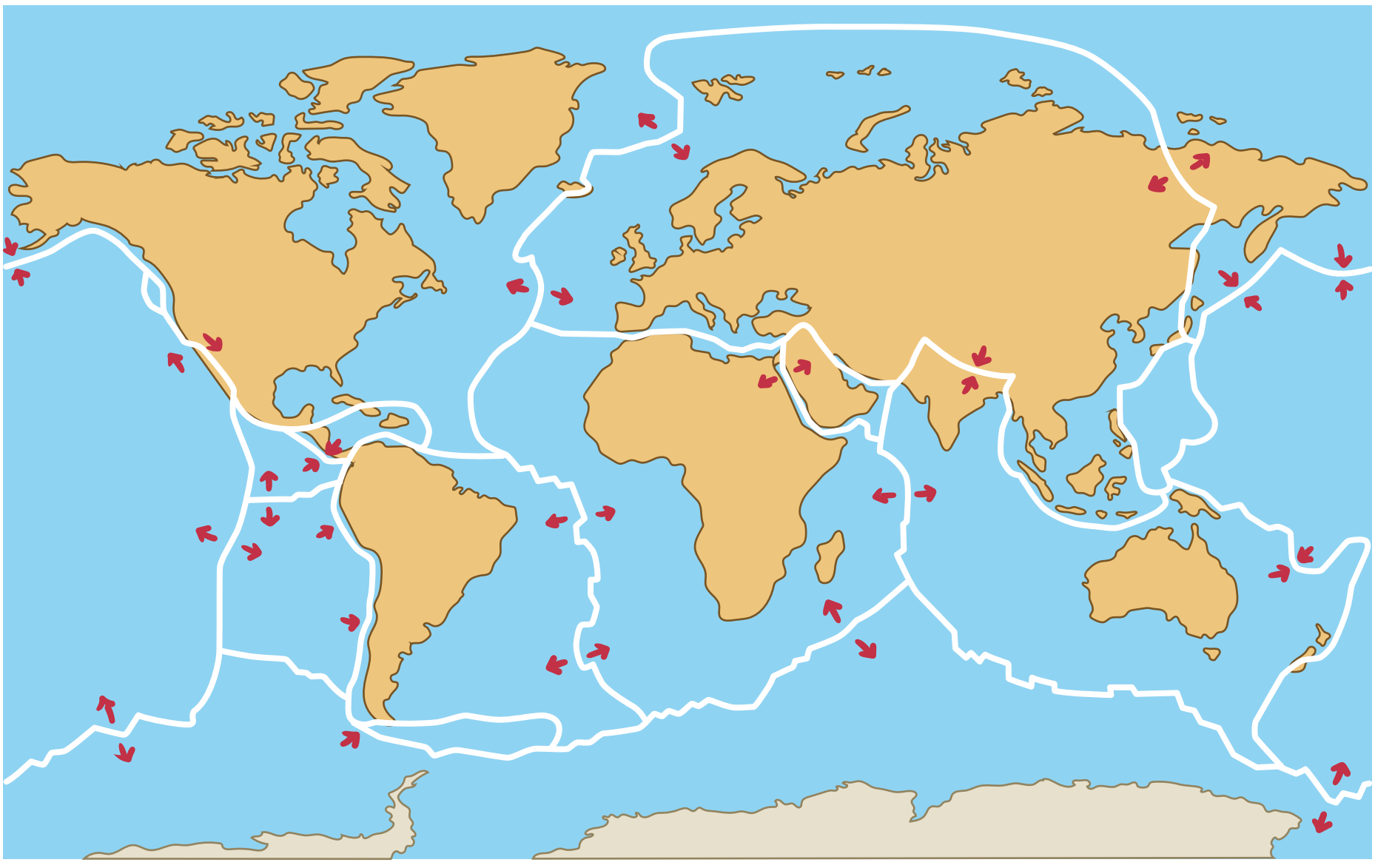
ALFRED WEGENER (1912)

EVIDÊNCIAS

PANGEIA

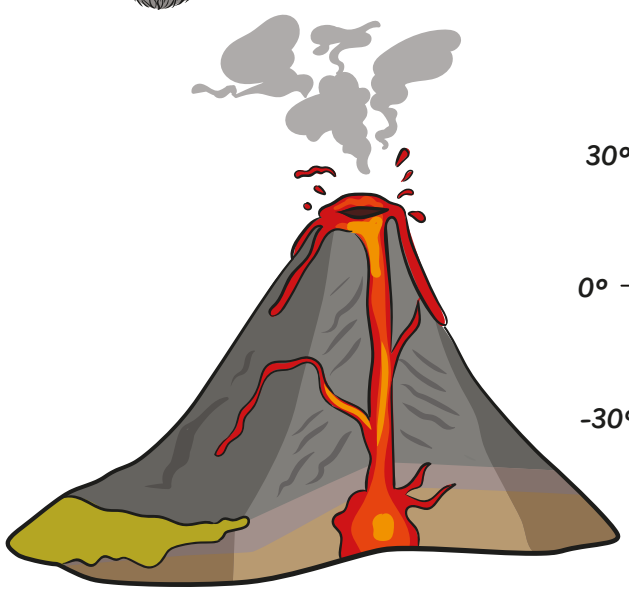


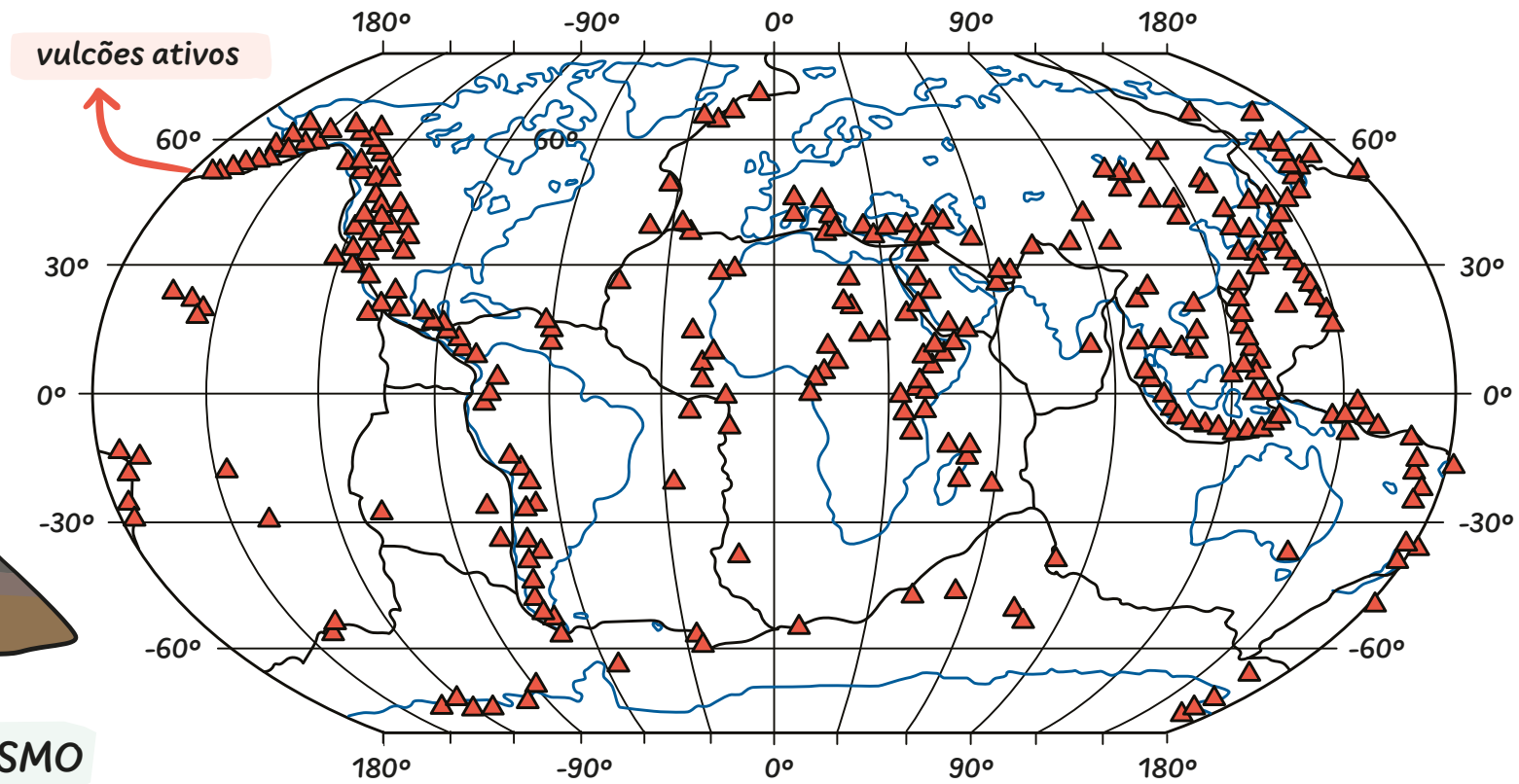
TECTÔNICA DE PLACAS



GEOGRAFIA
PROFESSOR GROTH

AGENTES ENDÓGENOS
(forças internas)





TECTONISMO - VULCANISMO

TECTONISMO OU DIASTROFISMO

movimento das placas tectônicas

→ **EPIROGÊNESE** — movimento vertical

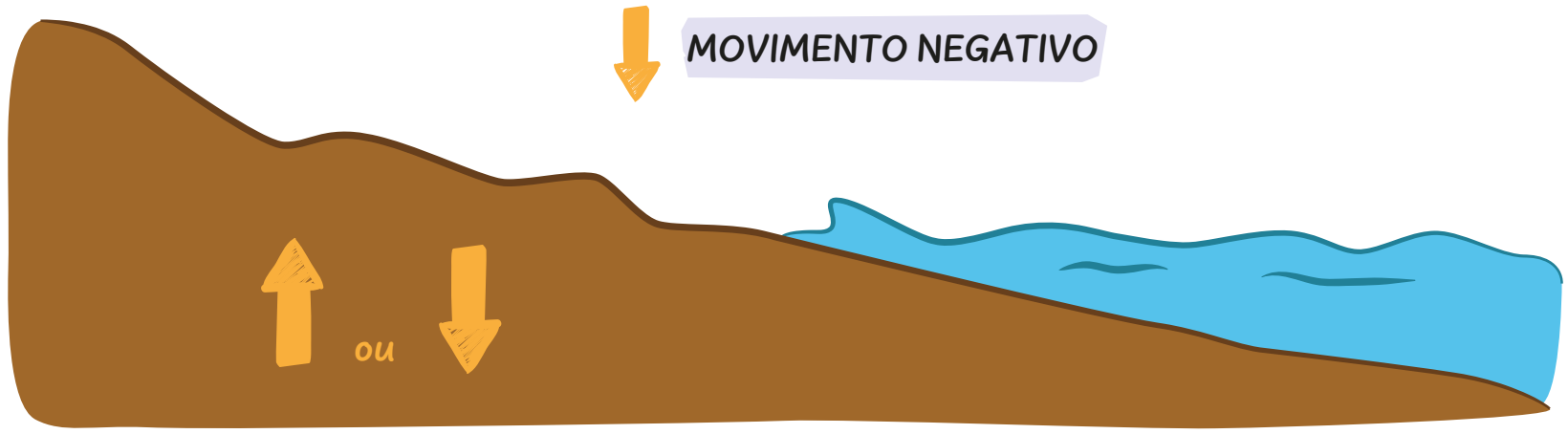
→ **OROGÊNESE** — movimento horizontal

EPIROGÊNESE

MOVIMENTO VERTICAL

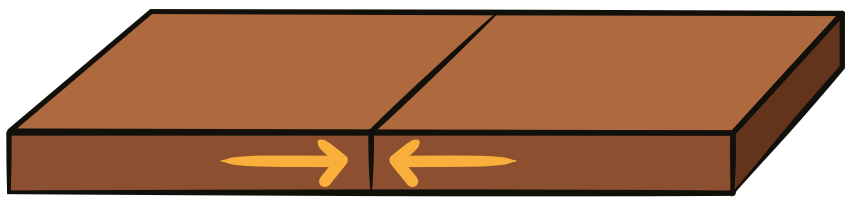
↑ MOVIMENTO POSITIVO

↓ MOVIMENTO NEGATIVO



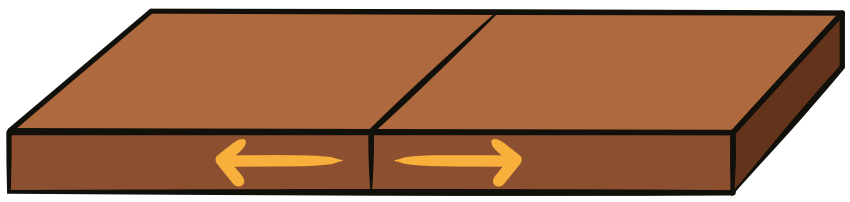
OROGÊNESE

CONVERGENTE (destrutivo)



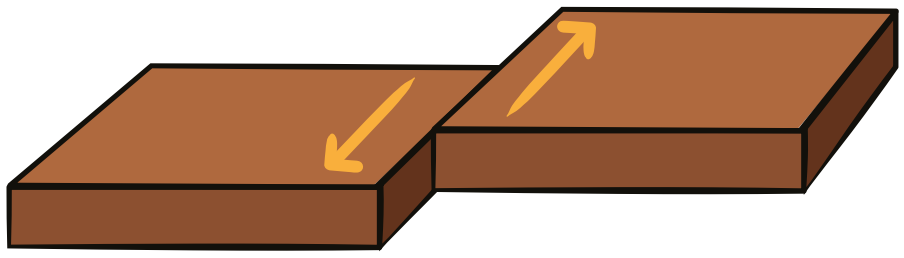
- Dobramentos modernos
- Fossas Oceânicas

DIVERGENTE (construtivo)



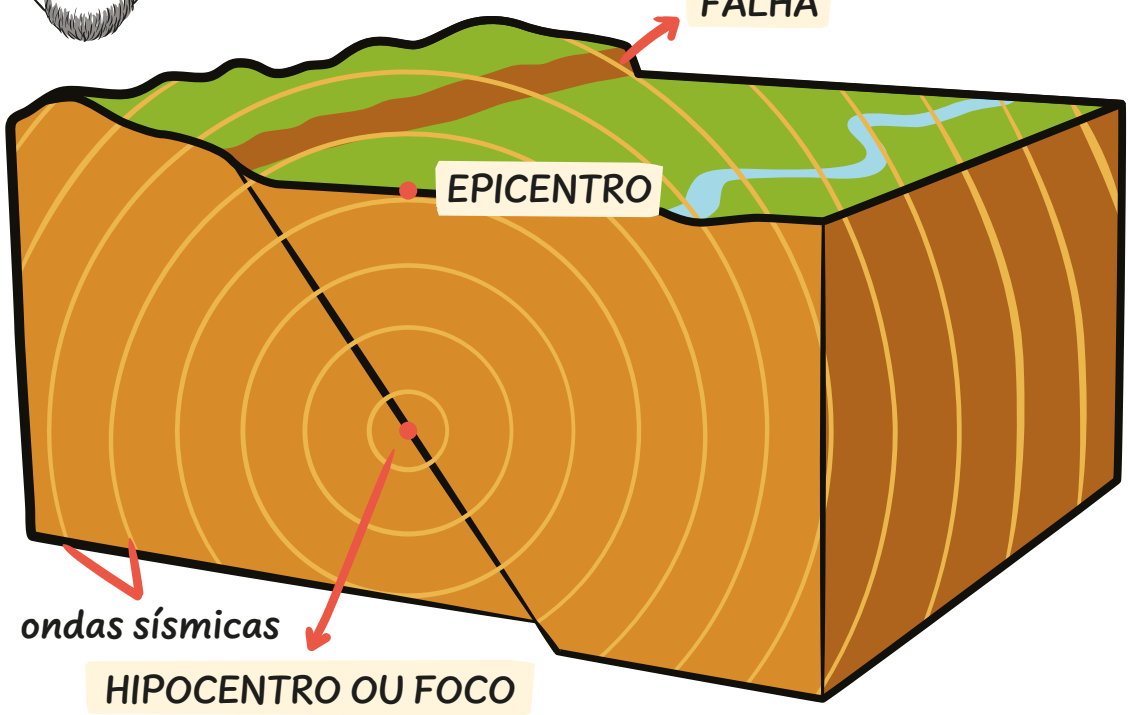
- Dorsais Oceânicas

TRANSFORMANTE ou TANGENCIAL (conservativo)



- Falha de San Andreas

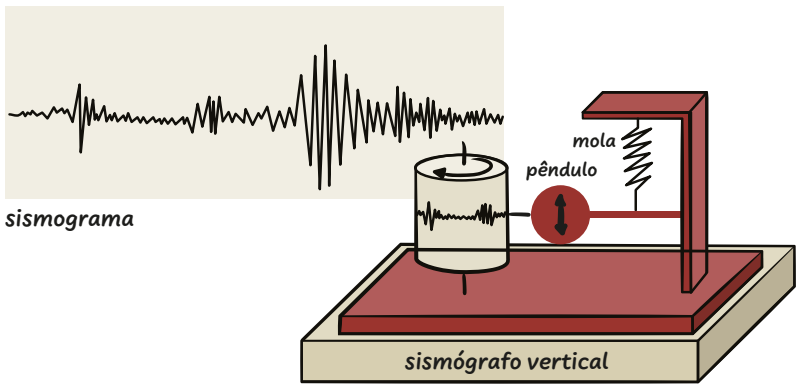
ABALOS SÍSMICOS, TERREMOTOS OU SISMOS



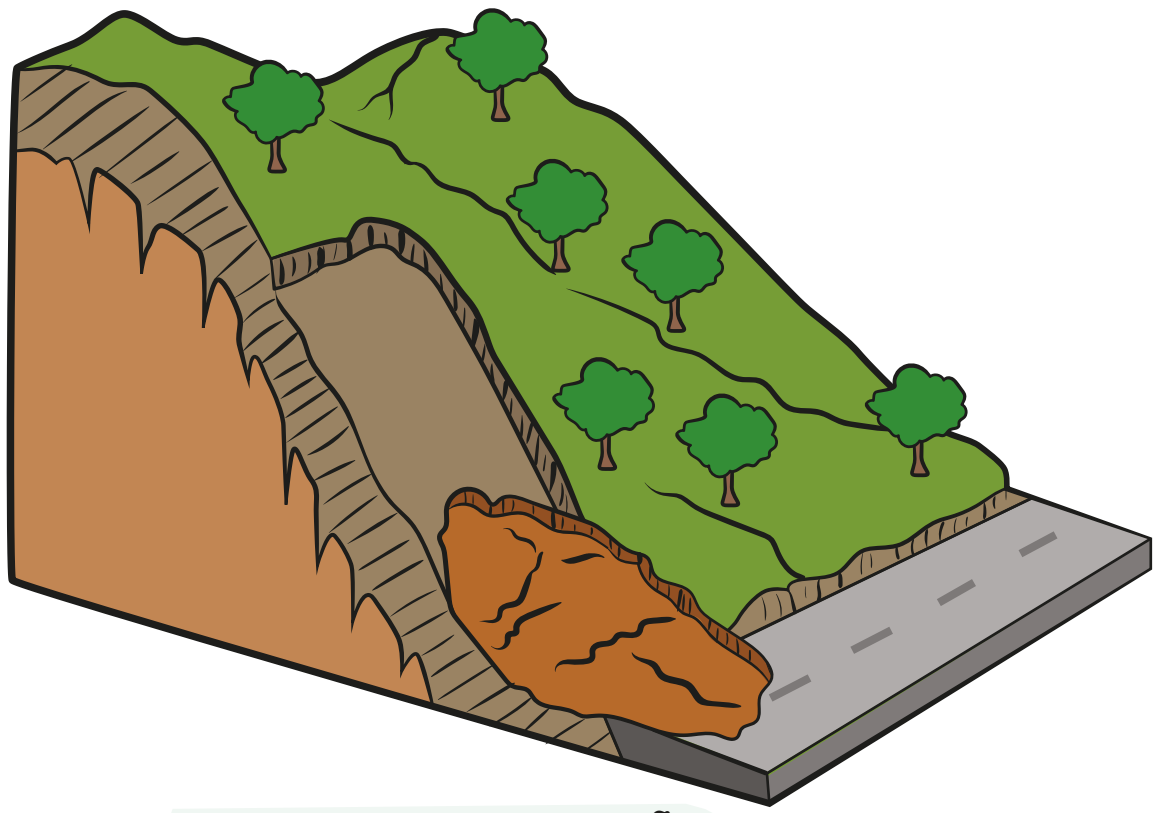
ESCALA RICHTER

MAGNITUDE DO SISMO

ESCALA LOGARÍTMICA



AGENTES EXÓGENOS
(forças externas)



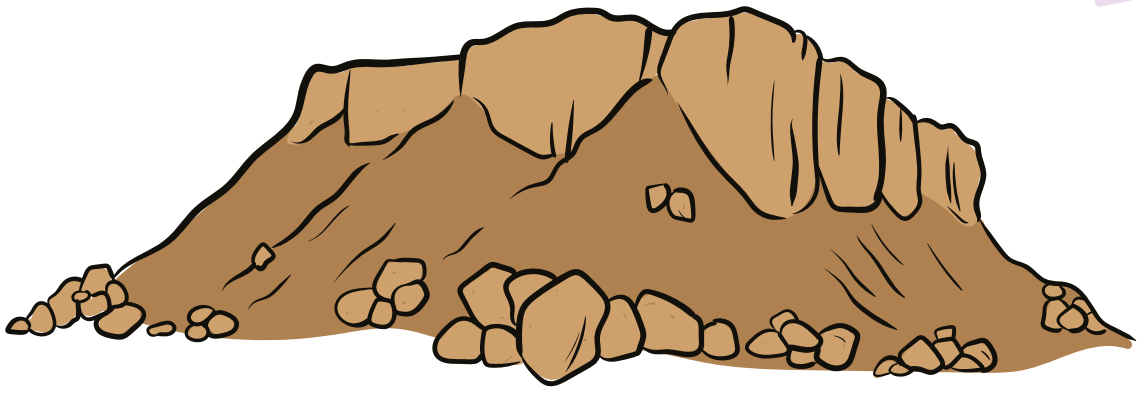
INTEMPERISMO - EROSÃO

INTEMPERISMO ou METEORIZAÇÃO

FÍSICO

QUÍMICO

BIOLÓGICO



Processo de DESGASTE / DESTRUIÇÃO das ROCHAS

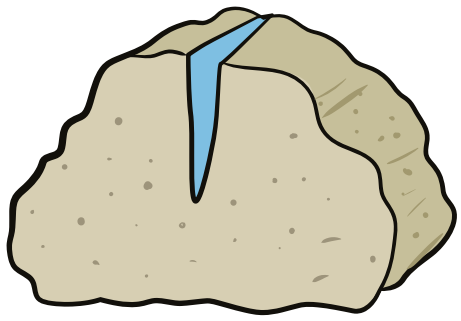
INTEMPERISMO FÍSICO

Processo de DESAGREGAÇÃO MECÂNICA da rocha

INTEMPERISMO QUÍMICO

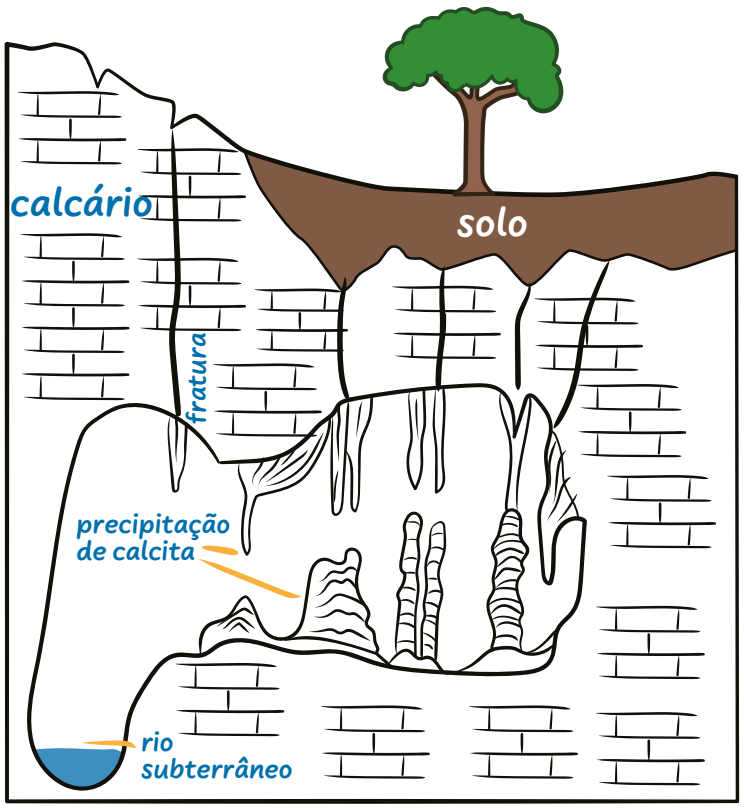
Processo de DECOMPOSIÇÃO QUÍMICA da rocha

AÇÃO DA ÁGUA

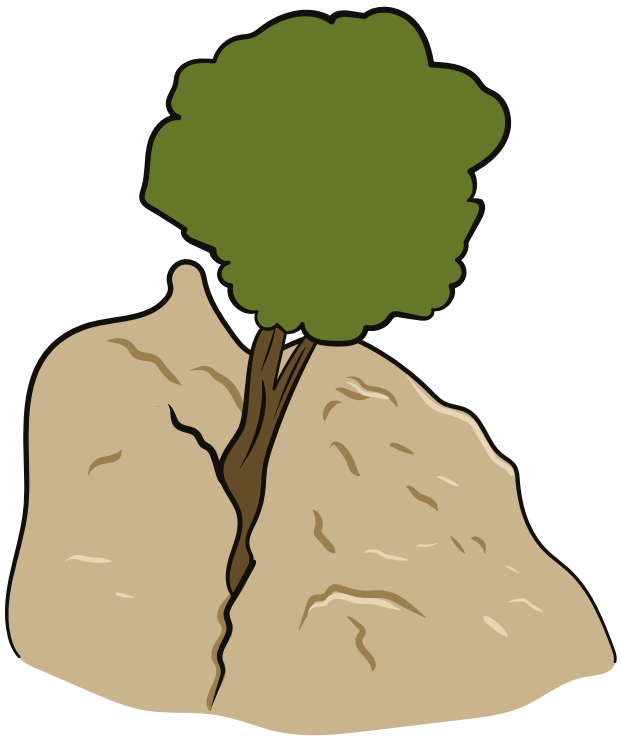


CLIMAS ÚMIDOS

Perfil cárstico



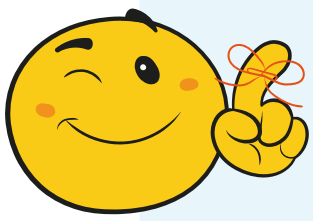
INTEMPERISMO ou METEORIZAÇÃO → FÍSICO / QUÍMICO → BIOLÓGICO



Ação química ou física dos seres vivos

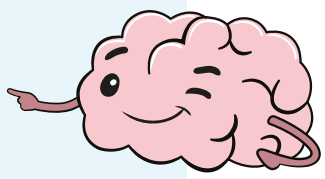
INTEMPERISMO ou METEORIZAÇÃO

Processo de DESGASTE / DESTRUIÇÃO das ROCHAS.



≠

EROSÃO



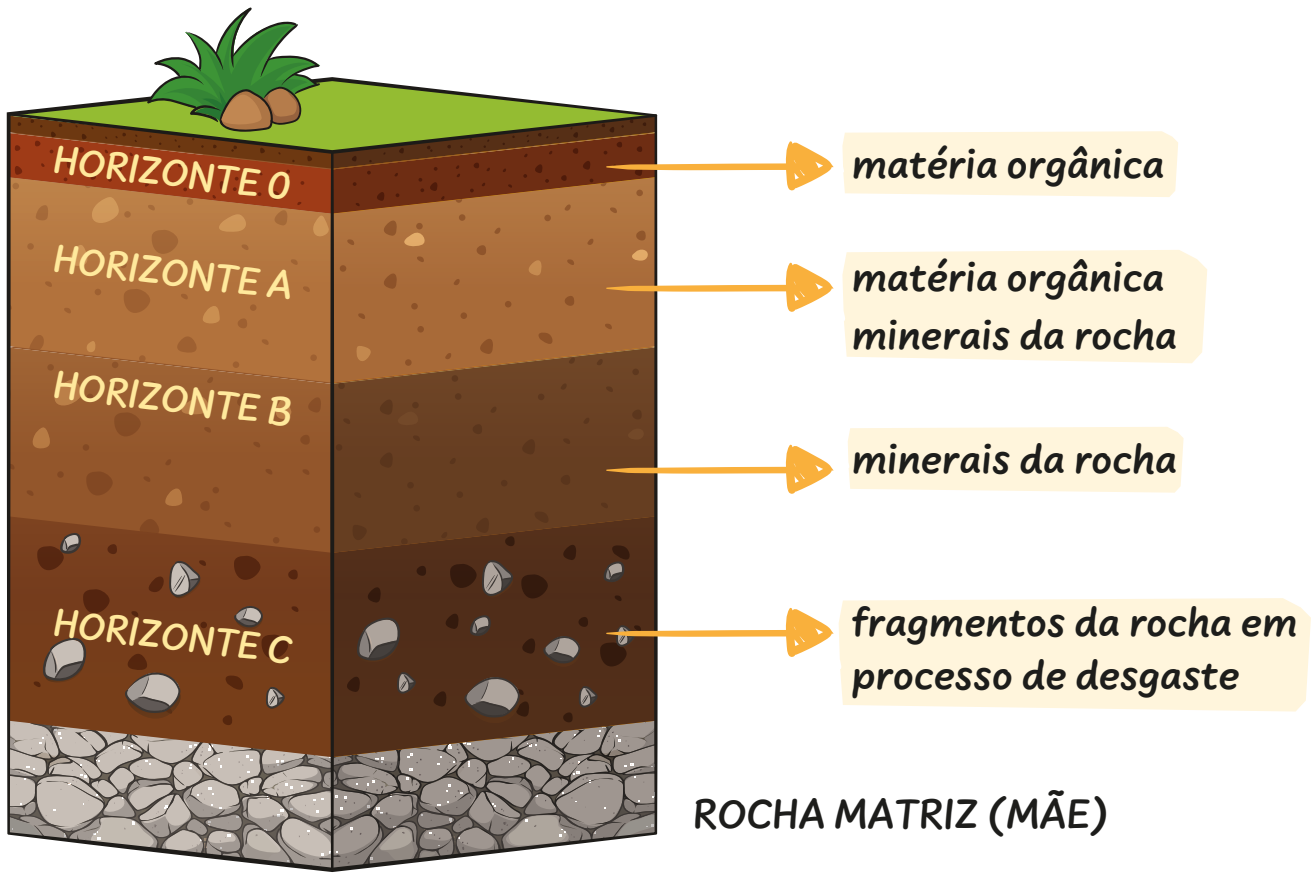
O material que sofreu intemperismo é desagregado e TRANSPORTADO (Chuva, rio, mar, vento e gelo).

PEDOGÊNESE — processo de formação do solo

→ INTEMPERISMO ou METEORIZAÇÃO das rochas

LIXIVIAÇÃO

Processo de extração ou solubilização seletiva de constituintes químicos pela ação da água.
Mais comum a dissolução e perda de Mg, Ca, K e Na.



EROSÃO

"Erosão é o conjunto de processos que promovem a retirada e transporte do material produzido pelo intemperismo, ocasionando o desgaste do relevo. Seus principais agentes são a água, o vento e o gelo."

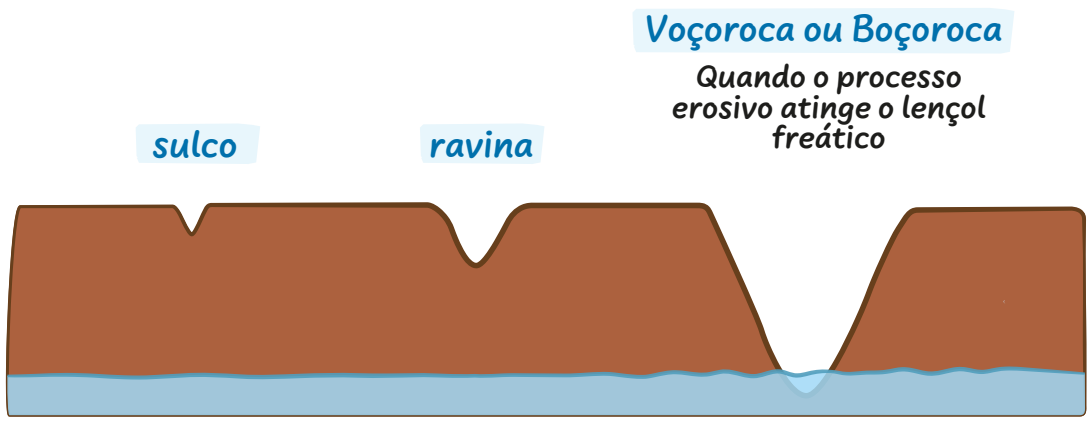
Fatores importantes no processo de EROSÃO :

Fonte: CPRM

- CLIMA - pluviosidade
- GEOLOGIA – tipo de solo
- GEOMORFOLOGIA - relevo
- AÇÃO ANTRÓPICA – desmatamento

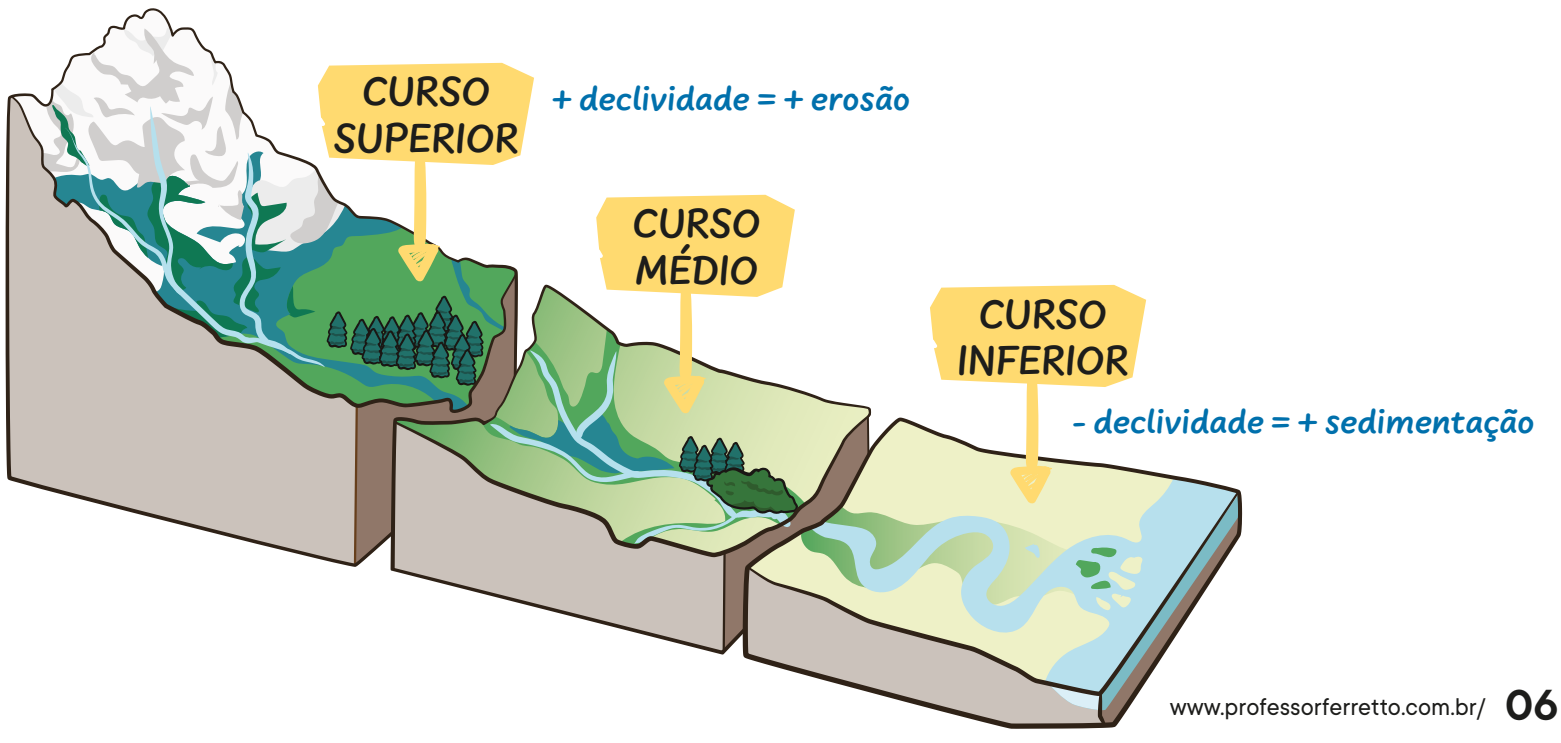
EROSÃO PLUVIAL

- Efeito splash ou salpicamento (gotejamento da chuva).
- Erosão laminar – perda superficial do solo pelo escoamento superficial da água.



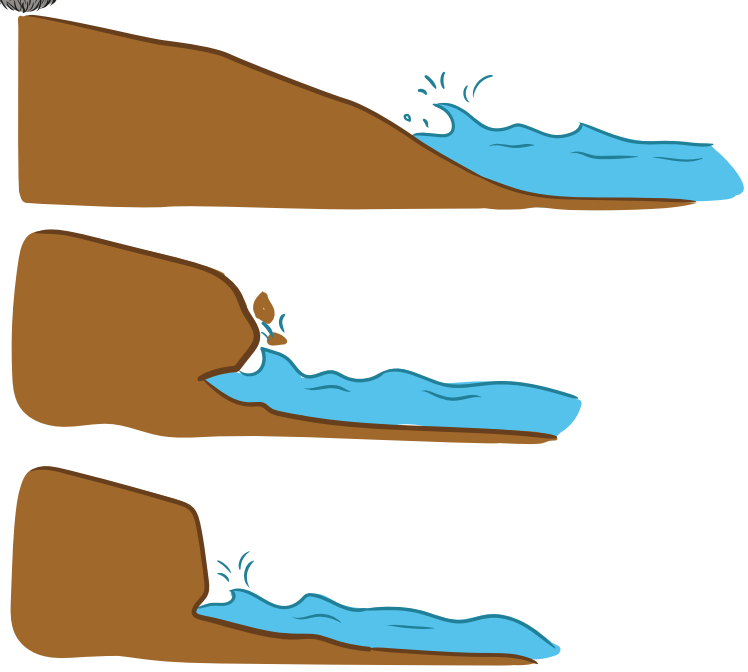
EROSÃO FLUVIAL (RIO)

PERFIL LONGITUDINAL

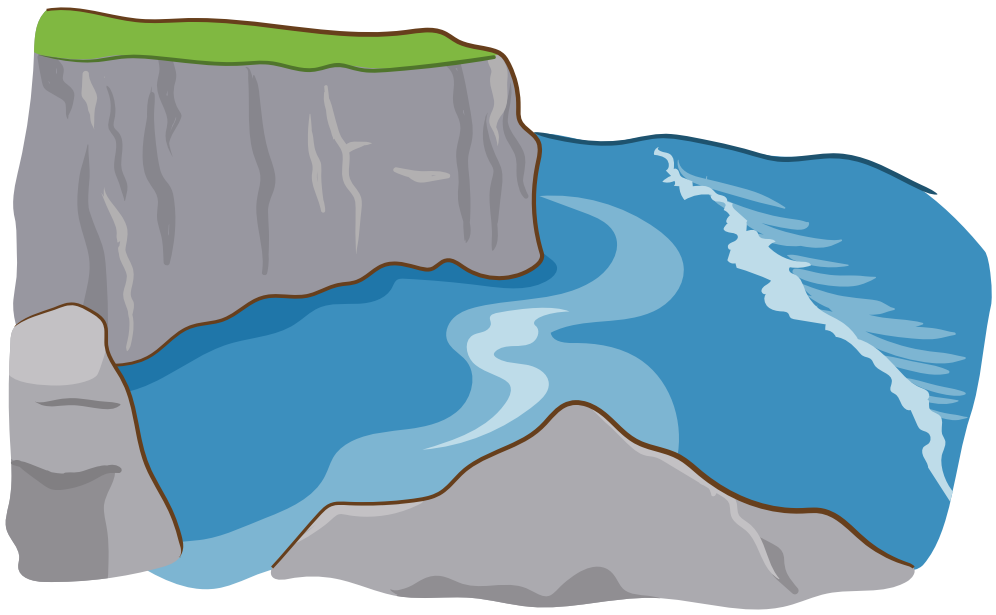


EROSÃO MARINHA (ABRASÃO)

FALÉSIAS

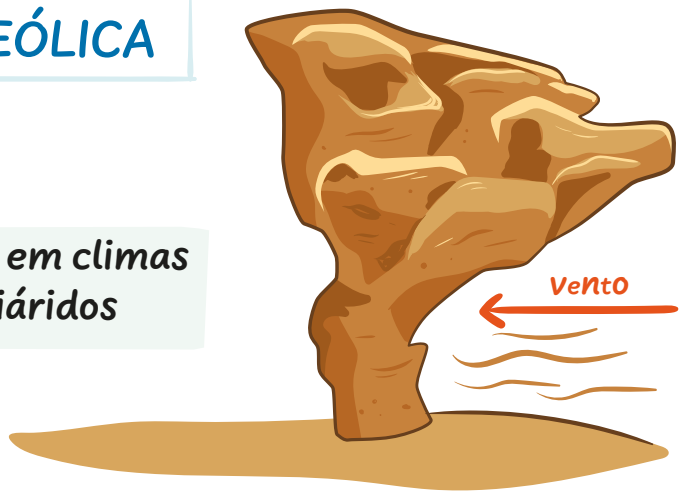


TORRES (RS)



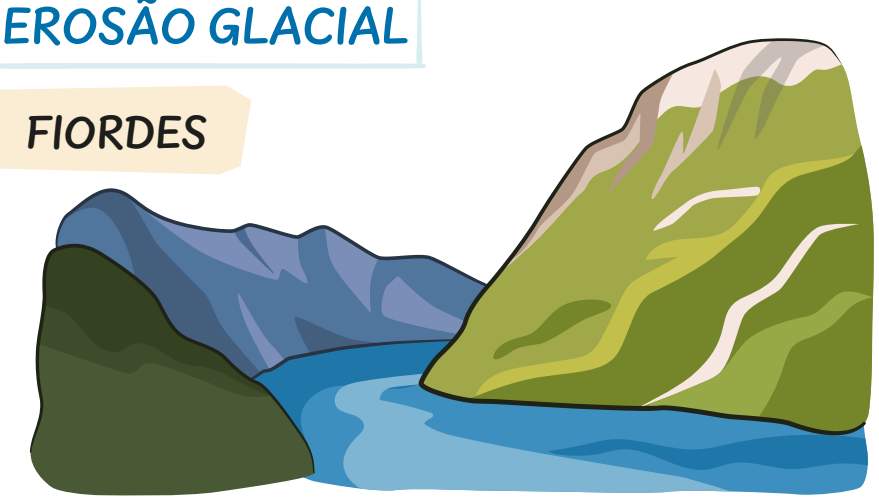
EROSÃO EÓLICA

Mais intensa em climas áridos e semiáridos

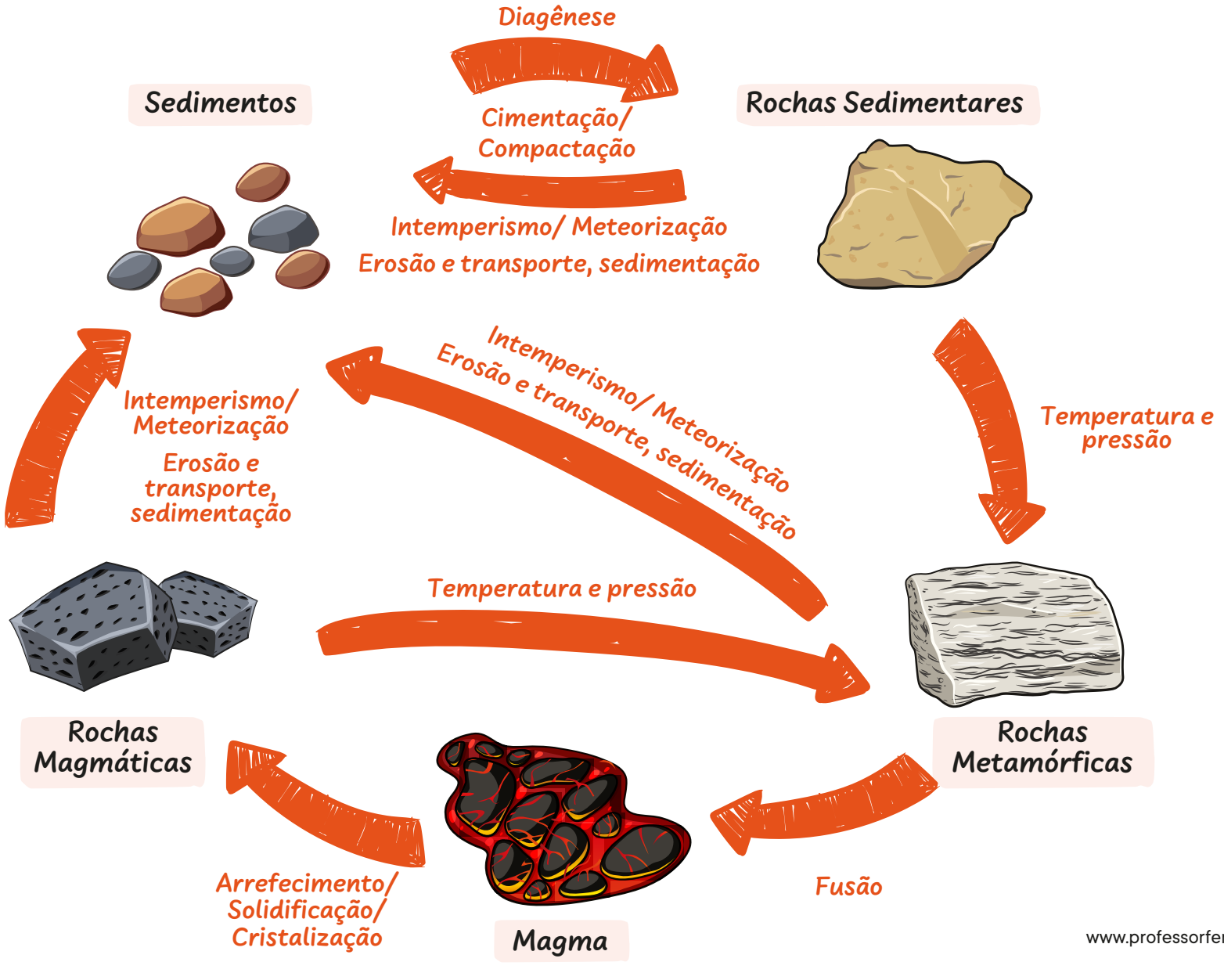


EROSÃO GLACIAL

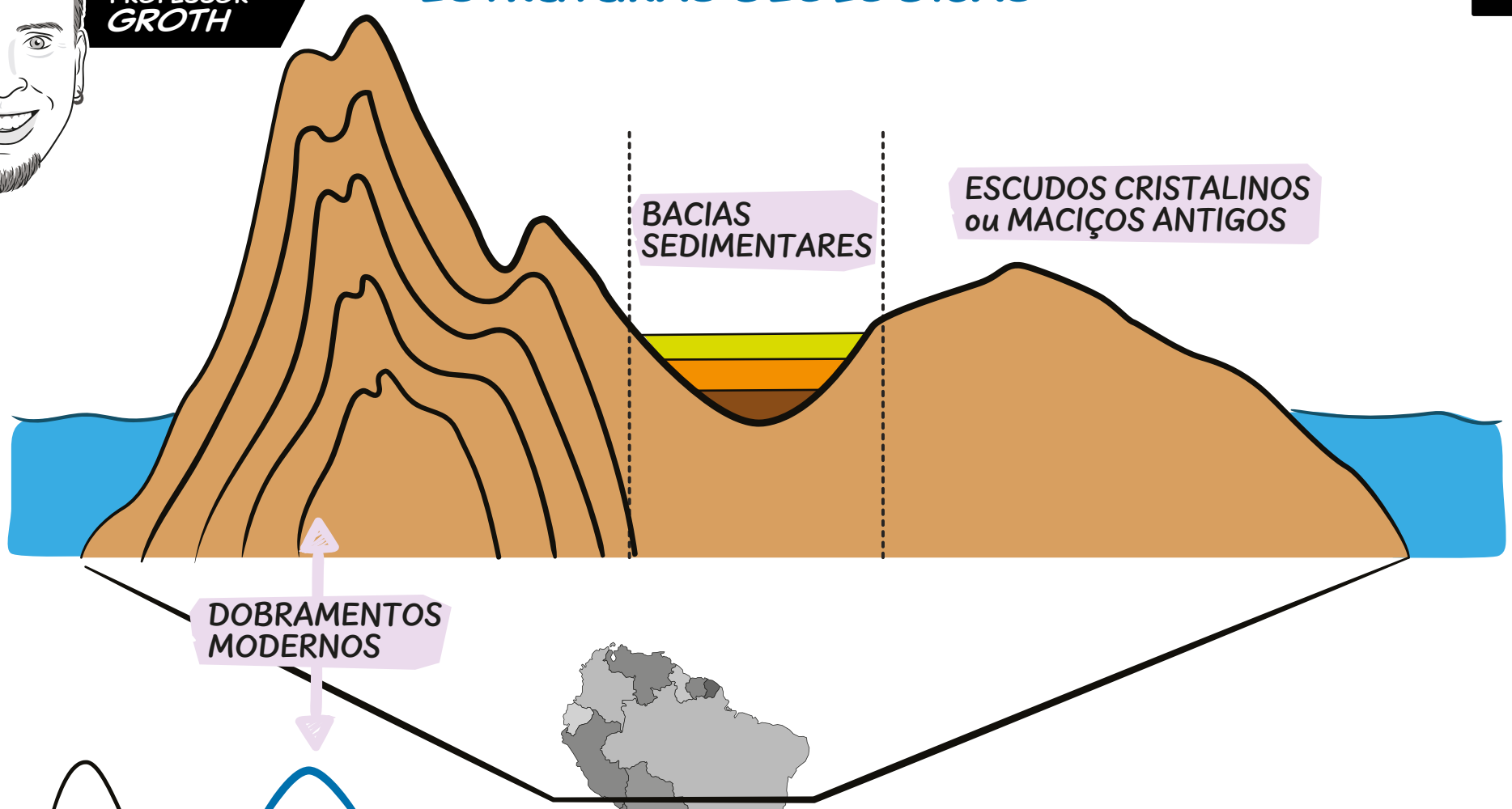
FIORDES

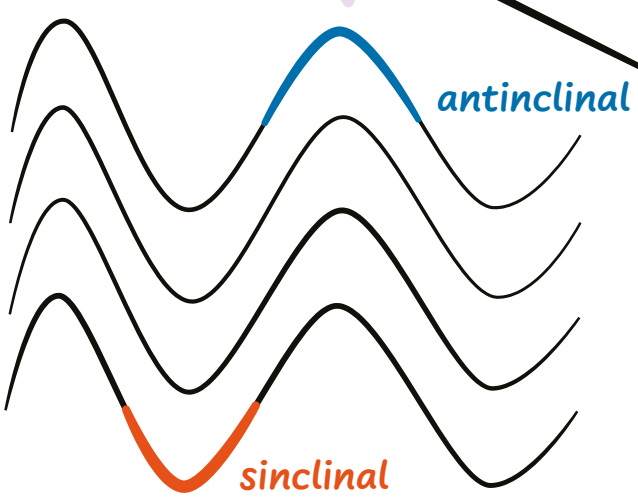


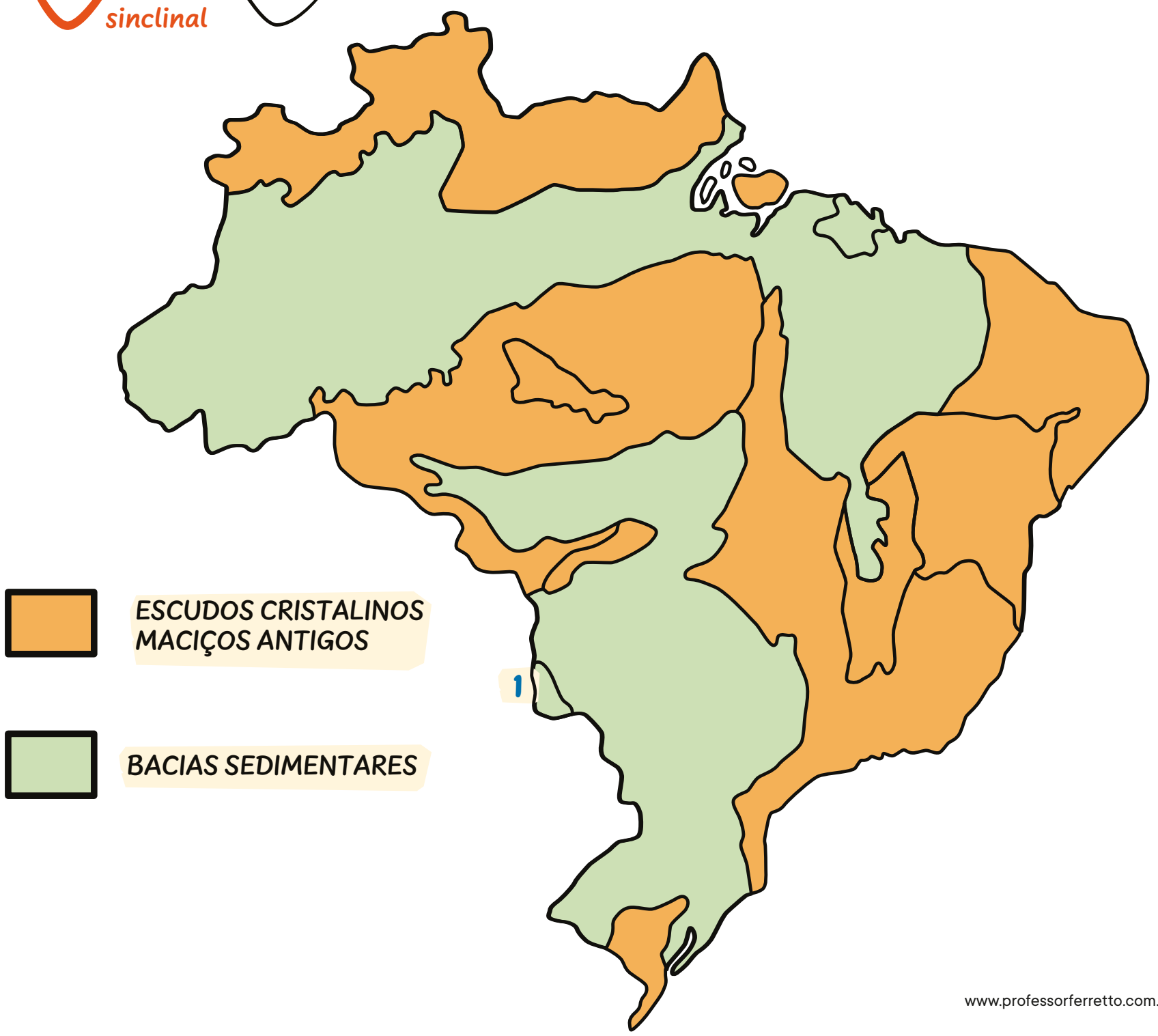
O CICLO DAS ROCHAS



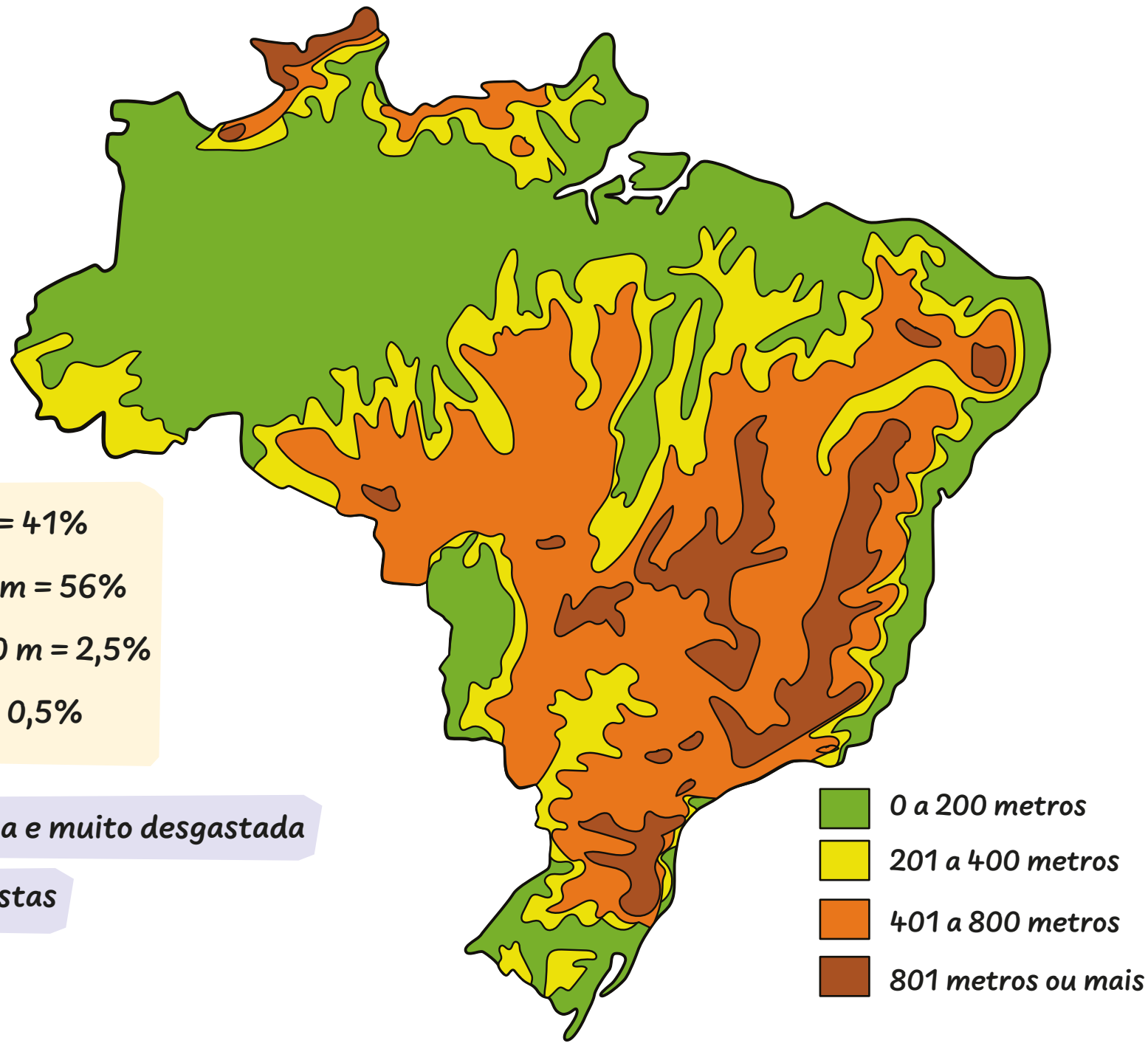
ESTRUTURAS GEOLÓGICAS







BRASIL MAPA HIPSOMÉTRICO



0 – 200 m = 41%

200 – 900 m = 56%

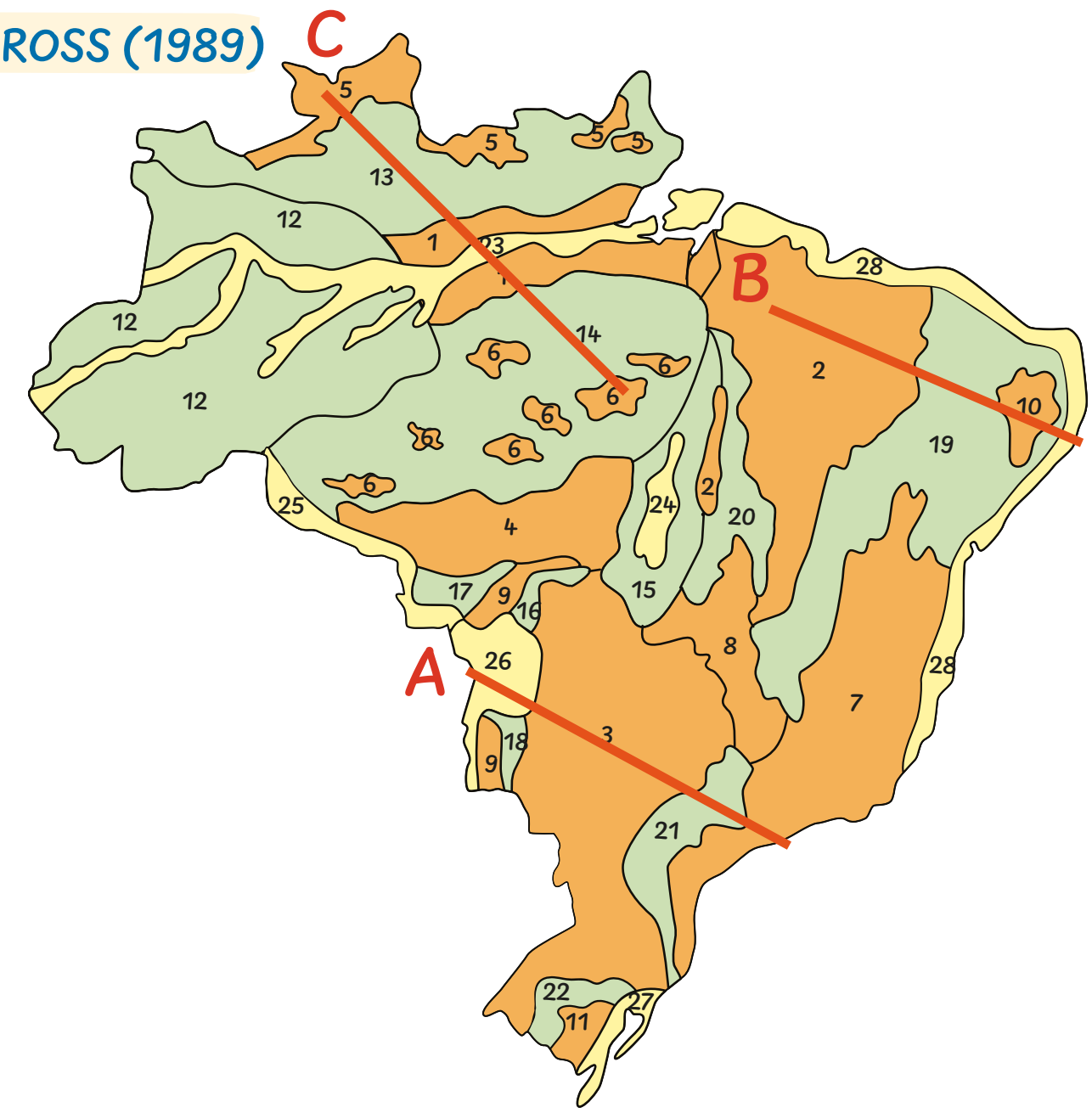
900 – 1200 m = 2,5%

+ 1200 m = 0,5%

- Formação antiga e muito desgastada
- Altitudes modestas

RELEVO BRASILEIRO

JURANDYR ROSS (1989)



F

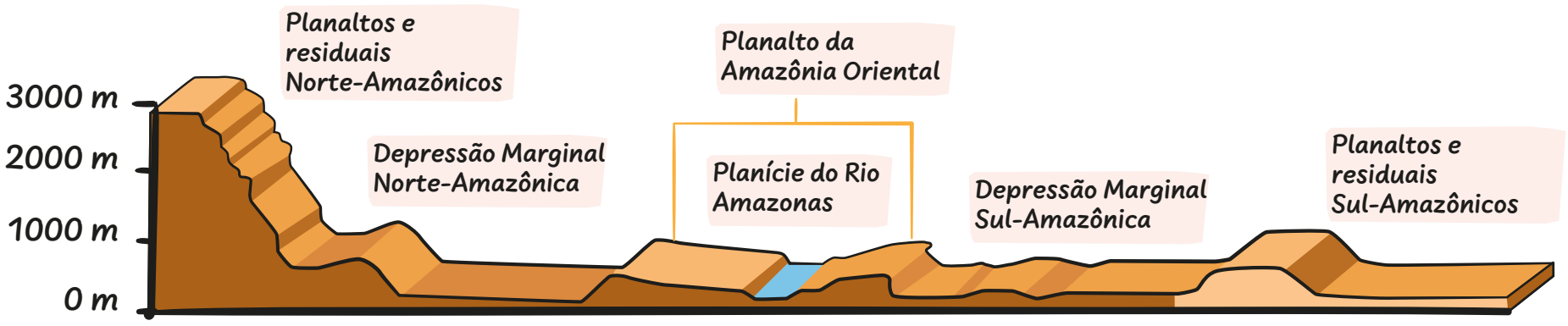
A

3000 m
2000 m
1000 m
0 m
Planície do Pantanal Mato-Grossense
Planaltos e chapadas da bacia do Paraná
Rio Paraná
Depressão periférica da borda leste da bacia do Paraná
Planaltos e serras do Atlântico leste-sudeste
Oceano Atlântico

B

3000 m
2000 m
1000 m
0 m
Rio Parnaíba
Planaltos e chapadas da bacia do rio Parnaíba
Escarpa (ex-serra) do Ibiapaba
Depressão sertaneja
Planalto da Borborema
Tabuleiros litorâneos
Oceano Atlântico

C


3000 m
2000 m
1000 m
0 m
Planaltos e residuais Norte-Amazônicos
Depressão Marginal Norte-Amazônica
Planalto da Amazônia Oriental
Planície do Rio Amazonas
Depressão Marginal Sul-Amazônica
Planaltos e residuais Sul-Amazônicos